Genuine Multipartite Entanglement Analysis — Molecular System

MOLECULE

Benzene (C6H6)
Basis set   : cc-pVDZ
Charge   : 0
Multiplicity : 1
Total e-   : 42
n-body order : 1

GROUND STATE ENERGIES

E0 (Ha) : -230.75355805
E1 (Ha) : -230.57826157
Gap (Ha) : 0.17529648
HF (Ha) : -230.71878874

CONVEX-ROOF METHODS

Methods run   : gcg
Restarts   : 3

C_LB : coherence witness bound
C_expo: unitary opt (exp map)
C_stf : Stiefel manifold (Alg. B)
C_UB : eigendecomposition bound

ACTIVE SPACE CAS(6,6)

Frozen core orbs : 18
Frozen e-   : 36
Active e-   : 6
Active orbs   : 6
Spin-orbitals   : 12
Hilbert dim   : 400

SPIN PROPERTIES

<S2>   : 0.000000
S   : 0.0000
<Sz>   : 0.000000
Multiplicity : 1
Spin state   : Singlet (S=0)

GME CRITERIA

C_GME = sqrt(2 * min_A (1 - Tr(rho_A^2)))

Convex roof (mixed):
  C_GME(rho) = min Sum p_i C_GME(|psi_i>)

Bounds ordering:
  C_LB <= C_GME <= C_UB

DOUBLE OCCUPANCY

<D> total : 2.961506
<D> / site : 0.493584

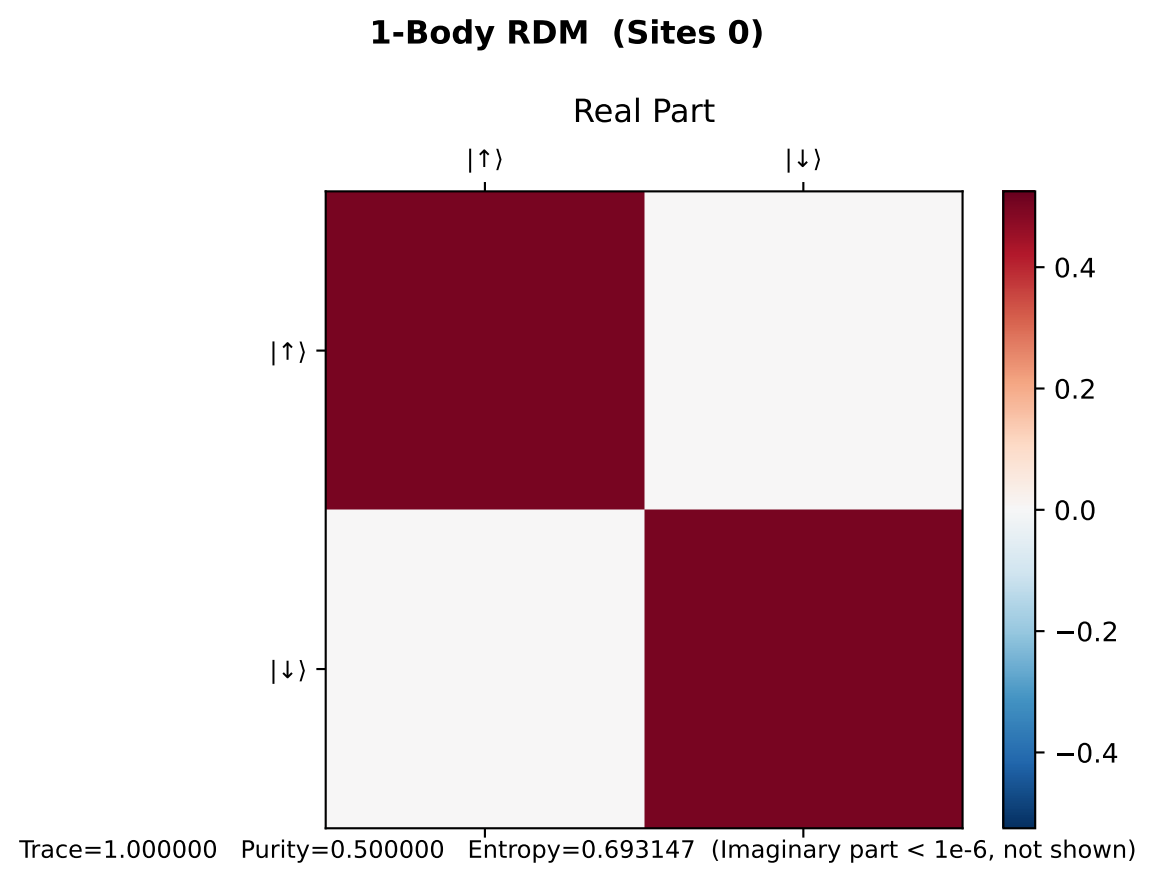
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Body order : 1
Clusters   : 6

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Summary — 1-Body RDMs (Benzene, CAS(6,6))

Orbitals	Purity	C_GME (pure)	C_LB	C_gcg	C_stiefel	C_UB	State
0	0.500000	—	—	—	—	—	WEAKLY MIXED
1	0.500000	—	—	—	—	—	MIXED
2	0.500000	—	—	—	—	—	WEAKLY MIXED
3	0.500000	—	—	—	—	—	MIXED
4	0.500000	—	—	—	—	—	MIXED
5	0.500000	—	—	—	—	—	MIXED


1-Body RDM (Sites 0)
Real Part
|↑⟩
|↓⟩
|↑⟩
|↓⟩
0.4
0.2
0.0
−0.2
−0.4
Trace=1.000000 Purity=0.500000 Entropy=0.693147 (Imaginary part < 1e-6, not shown)

Analysis Summary - 1-Body RDM (Sites 0)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.5000000000
Von Neumann entropy	: 0.6931471806
Eigenvalue min	: 5.000000e-01
Eigenvalue max	: 5.000000e-01
Imaginary part	: < 1e-6 (not shown)

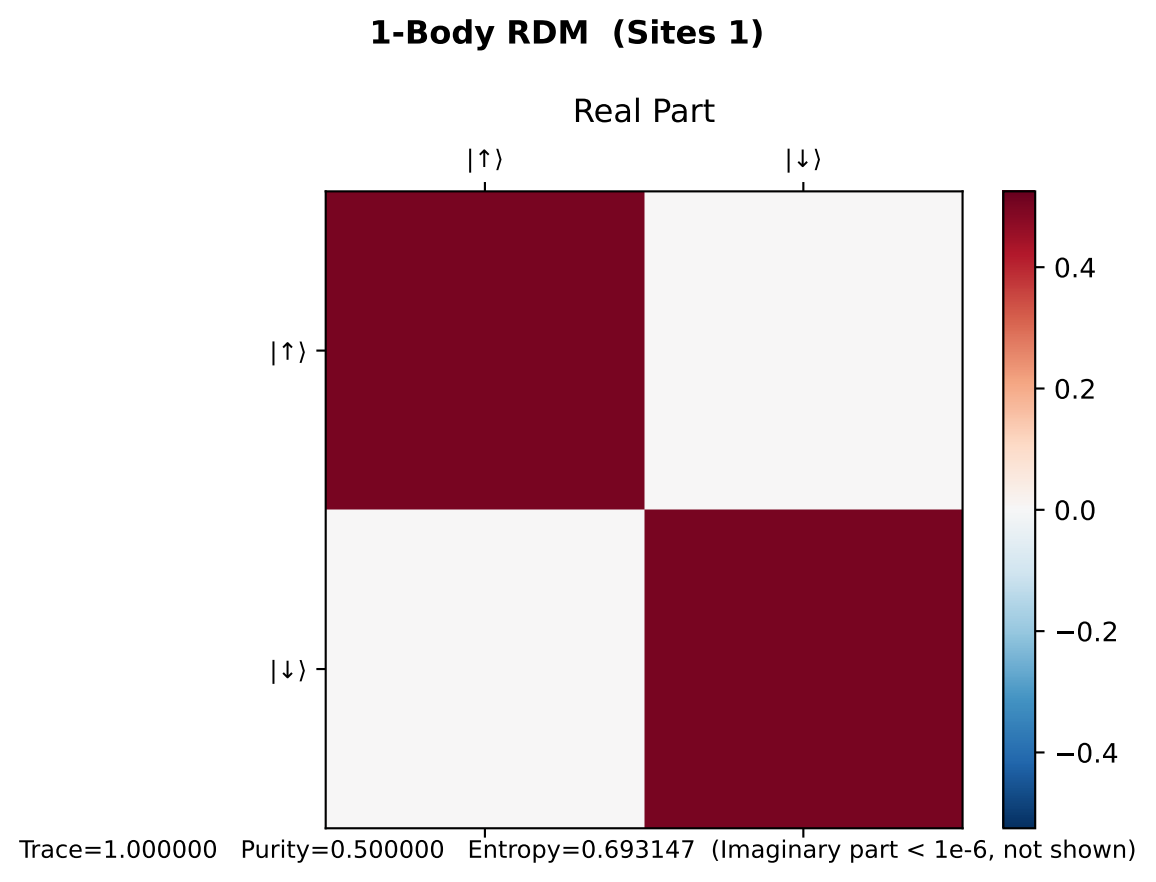
GME Analysis
==========

gamma = Tr(rho^2) = 0.50000000

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


1-Body RDM (Sites 1)
Real Part
|↑⟩
|↓⟩
|↑⟩
|↓⟩
0.4
0.2
0.0
−0.2
−0.4
Trace=1.000000 Purity=0.500000 Entropy=0.693147 (Imaginary part < 1e-6, not shown)

Analysis Summary - 1-Body RDM (Sites 1)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.5000000000
Von Neumann entropy	: 0.6931471806
Eigenvalue min	: 5.000000e-01
Eigenvalue max	: 5.000000e-01
Imaginary part	: < 1e-6 (not shown)

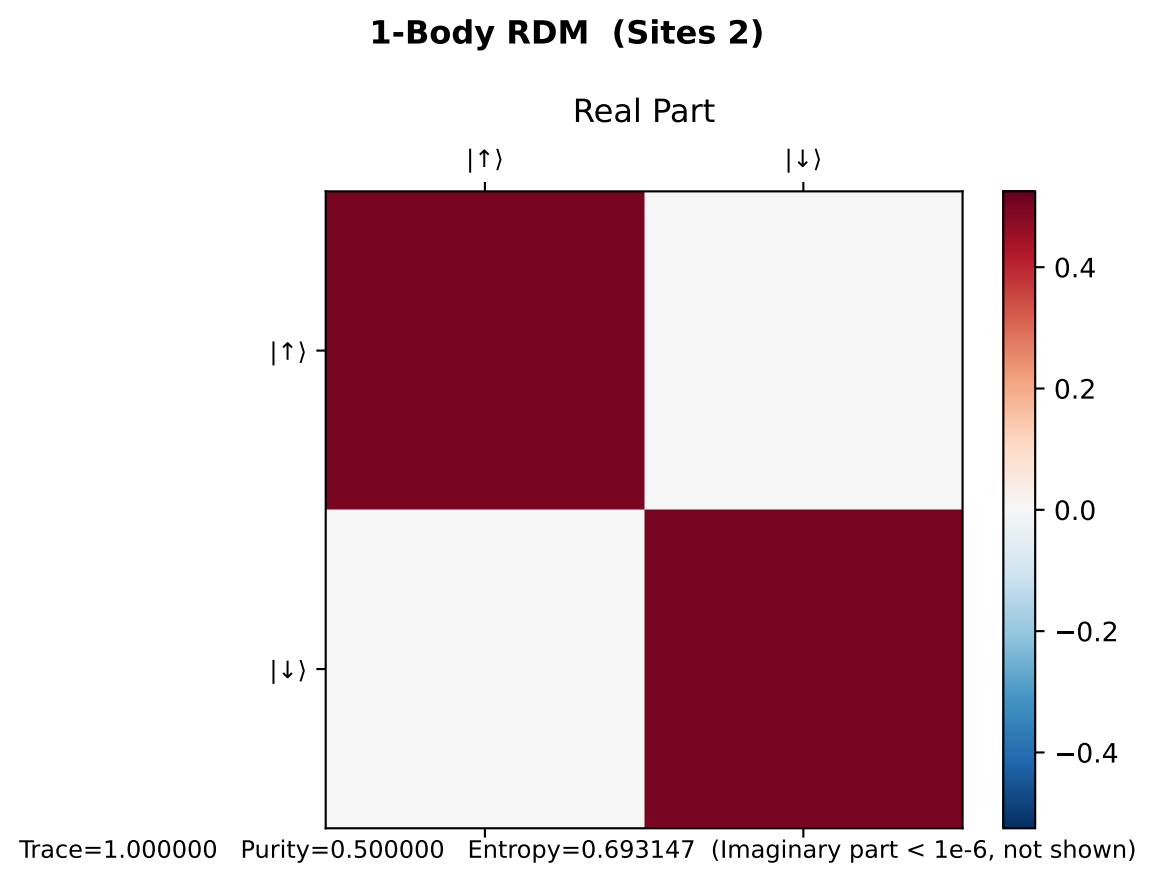
GME Analysis
==========

gamma = Tr(rho^2) = 0.50000000

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


1-Body RDM (Sites 2)
Real Part
|↑⟩
|↓⟩
|↑⟩
|↓⟩
0.4
0.2
0.0
−0.2
−0.4
Trace=1.000000 Purity=0.500000 Entropy=0.693147 (Imaginary part < 1e-6, not shown)

Analysis Summary - 1-Body RDM (Sites 2)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.5000000000
Von Neumann entropy	: 0.6931471806
Eigenvalue min	: 5.000000e-01
Eigenvalue max	: 5.000000e-01
Imaginary part	: < 1e-6 (not shown)

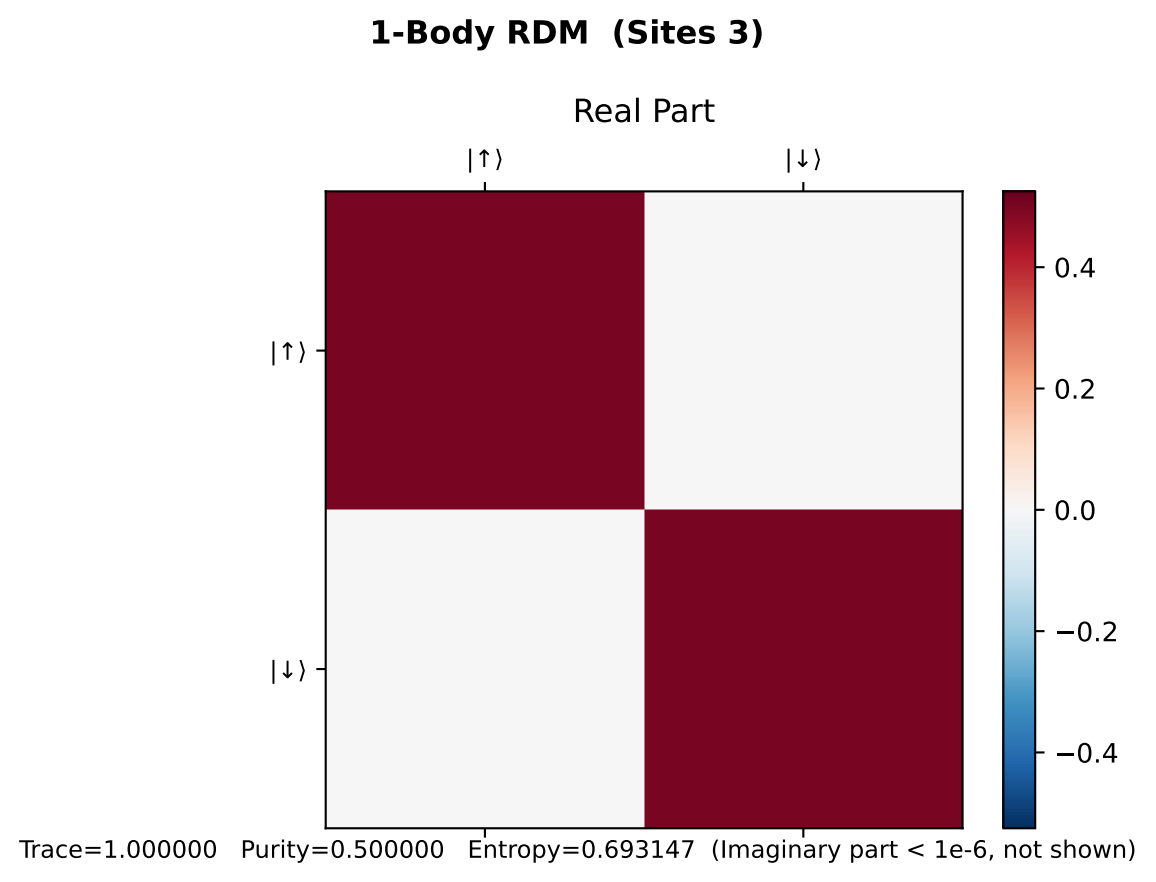
GME Analysis
==========

gamma = Tr(rho^2) = 0.50000000

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


1-Body RDM (Sites 3)
Real Part
|↑⟩
|↓⟩
|↑⟩
|↓⟩
0.4
0.2
0.0
−0.2
−0.4
Trace=1.000000 Purity=0.500000 Entropy=0.693147 (Imaginary part < 1e-6, not shown)

Analysis Summary - 1-Body RDM (Sites 3)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.5000000000
Von Neumann entropy	: 0.6931471806
Eigenvalue min	: 5.000000e-01
Eigenvalue max	: 5.000000e-01
Imaginary part	: < 1e-6 (not shown)

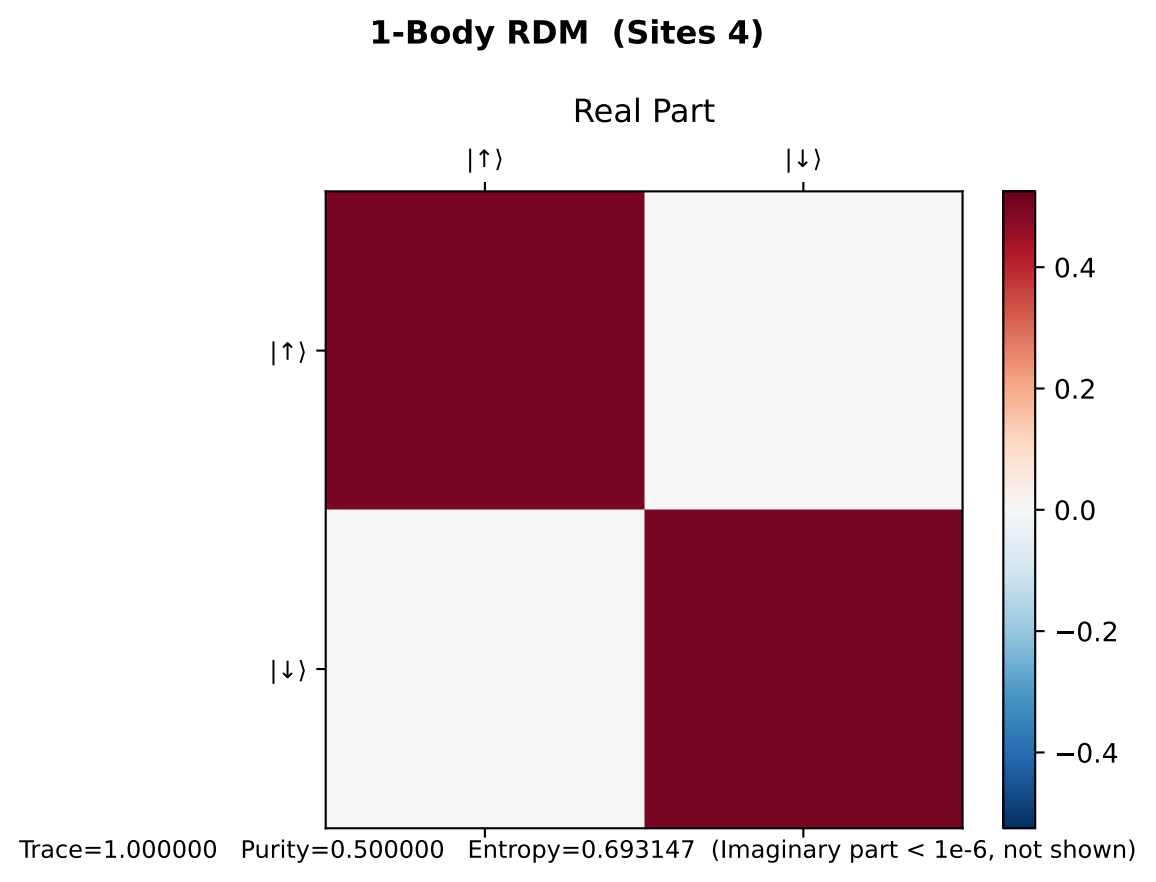
GME Analysis
==========

gamma = Tr(rho^2) = 0.50000000

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


1-Body RDM (Sites 4)
Real Part
|↑⟩
|↓⟩
|↑⟩
|↓⟩
0.4
0.2
0.0
−0.2
−0.4
Trace=1.000000 Purity=0.500000 Entropy=0.693147 (Imaginary part < 1e-6, not shown)

Analysis Summary - 1-Body RDM (Sites 4)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.5000000000
Von Neumann entropy	: 0.6931471806
Eigenvalue min	: 5.000000e-01
Eigenvalue max	: 5.000000e-01
Imaginary part	: < 1e-6 (not shown)

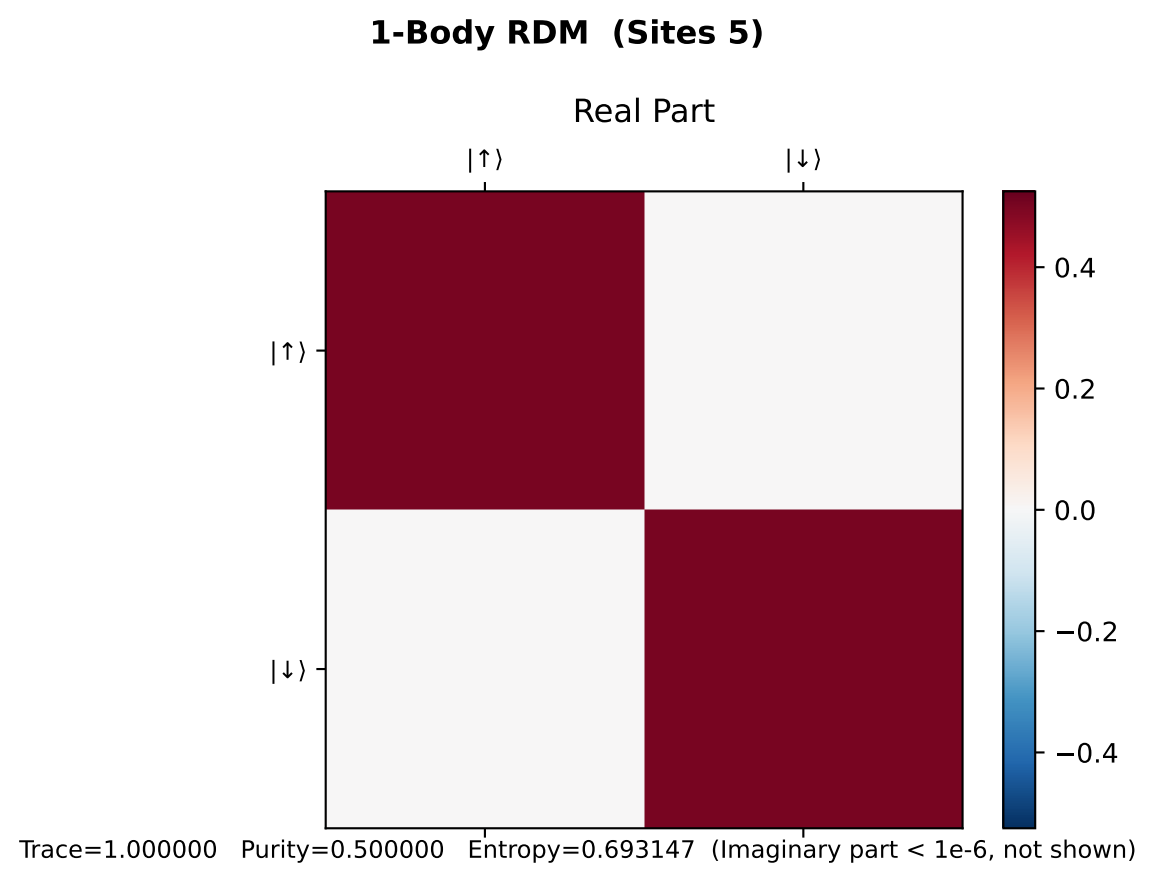
GME Analysis
==========

gamma = Tr(rho^2) = 0.50000000

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


1-Body RDM (Sites 5)
Real Part
|↑⟩
|↓⟩
|↑⟩
|↓⟩
0.4
0.2
0.0
−0.2
−0.4
Trace=1.000000 Purity=0.500000 Entropy=0.693147 (Imaginary part < 1e-6, not shown)

Analysis Summary - 1-Body RDM (Sites 5)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.5000000000
Von Neumann entropy	: 0.6931471806
Eigenvalue min	: 5.000000e-01
Eigenvalue max	: 5.000000e-01
Imaginary part	: < 1e-6 (not shown)

GME Analysis
==========

gamma = Tr(rho^2) = 0.50000000

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.